KU LEUVEN

FACULTY OF ENGINEERING TECHNOLOGY
MASTER OF ARTIFICIAL INTELLIGENCE IN BUSINESS AND INDUSTRY

# Image Classification for detection of Melanoma

Industrial Perspective Project - Barco

Ashok George
Muhammad Rafiq

November 2025

## 1. Introduction

The project for image classification is an introduction into the practical application of Machine Learning. It gives a sneak peak into use of python functions like pandas and plotting the graphical impact of the Machine learning.

**The objective here is to**

- Build a **deep learning model** to classify dermatoscopic images into **8 classes**
- ['bkl', 'nv', 'df', 'mel', 'vasc', 'bcc', 'akiec', 'other']
- Train the model on a **legacy dataset** and validate on a **target dataset**.
- Evaluate generalization performance on a **test dataset**.
- Analyze overfitting, class imbalance, and model performance via **metrics and confusion matrix**.

## 2. Running the code

The original code is run on google colab. However, since the code needs to be run multiple times and performance is throttled on google colab due to shared GPU for free users, we had to run on local machine. Hence the code we have uploaded is that from the output on running the code in local machine.

Data Sources:

1. **Training set**
   Images -> legacy
   Meta data -> metadata_legacy.csv

2. **Validation set**
   Images -> target
   Meta data -> metadata_target.csv

3. **Testing set**
   Images -> target
   Meta data -> metadata_target.csv

| Name | Date modified | Type | Size |
|---|---|---|---|
| Barco_project | 25-10-2025 20:01 | File folder | |
| code | 25-10-2025 18:17 | File folder | |
| docker | 25-10-2025 12:23 | File folder | |
| info | 25-10-2025 12:23 | File folder | |
| legacy | 12-11-2025 11:31 | File folder | |
| second_wave | 11-11-2025 22:50 | File folder | |
| target | 25-10-2025 12:25 | File folder | |
| test | 16-11-2025 14:19 | File folder | |
| unlabeled | 25-10-2025 12:21 | File folder | |
| metadata_legacy | 12-11-2025 11:32 | Microsoft Excel Co... | 471 KB |
| metadata_second_wave | 11-11-2025 22:38 | Microsoft Excel Co... | 239 KB |
| metadata_target | 25-10-2025 12:03 | Microsoft Excel Co... | 67 KB |
| metadata_test | 11-11-2025 22:38 | Microsoft Excel Co... | 23 KB |

The additional set of images under the folder “second_wave” was used to train the system.

## 3. **Changes made to the given code**

The original code given covers the broad spectrum of a basic machine learning algo for image classification. But what we observed is that it can be tweaked to improve its performance, i.e. accuracy.

a. **Training cycle increased from 10 to 30:** This will train more and hence should give more accuracy
b. **Activation function:** Originally activation function was resnet18. But for image classification, a more widely model is efficient50
c. **Weights for classes:** The size of data for training was not distributed uniformly. Hence to normalise it, weights have been given in the training process
d. **Labels in training data:** The given training data contain 7 types of diseases. Each type corresponds to a particular skin disease, of which some as malignant and some benign. There were spelling variations which led to incorrect classification. Hence that has been rectified.
e. **Test data:** We have trained and validated the system and then tested it separately on aa set of test data. This tests the model for real world performance.

## 4. **Measuring performance**

The accuracy of the system under validation data has been computed and apart from that we have also added a graph to denote the accuracy over test data and further confusion matrix has been plotted to know if the model overfits on the training data or not.
From the plot of validation vs training loss, we can see that the point of overfitting is not yet reached so adding more epoch will give better accuracy.

From the confusion matrix, we can see that it is very good at predicting bkl, nv, vasc & bcc. But at the same time, it is not very good at akiec, df and mel.
If we categorise **mel, bcc, akiec** as the life-threatening cases, the algorithm only predicts correctly the cancer as per the table below.

| Lesion category | Predicted correctly | Predicted incorrectly | Accuracy | Malignancy detected |
|---|---|---|---|---|
| Mel | 27 | 23 | 54% | 58% (29) |
| Bcc | 39 | 11 | 78% | 80% (40) |
| akiec | 1 | 49 | 2% | 40% (20) |

**Overall malignancy detection of the system currently stands at 59.3%**

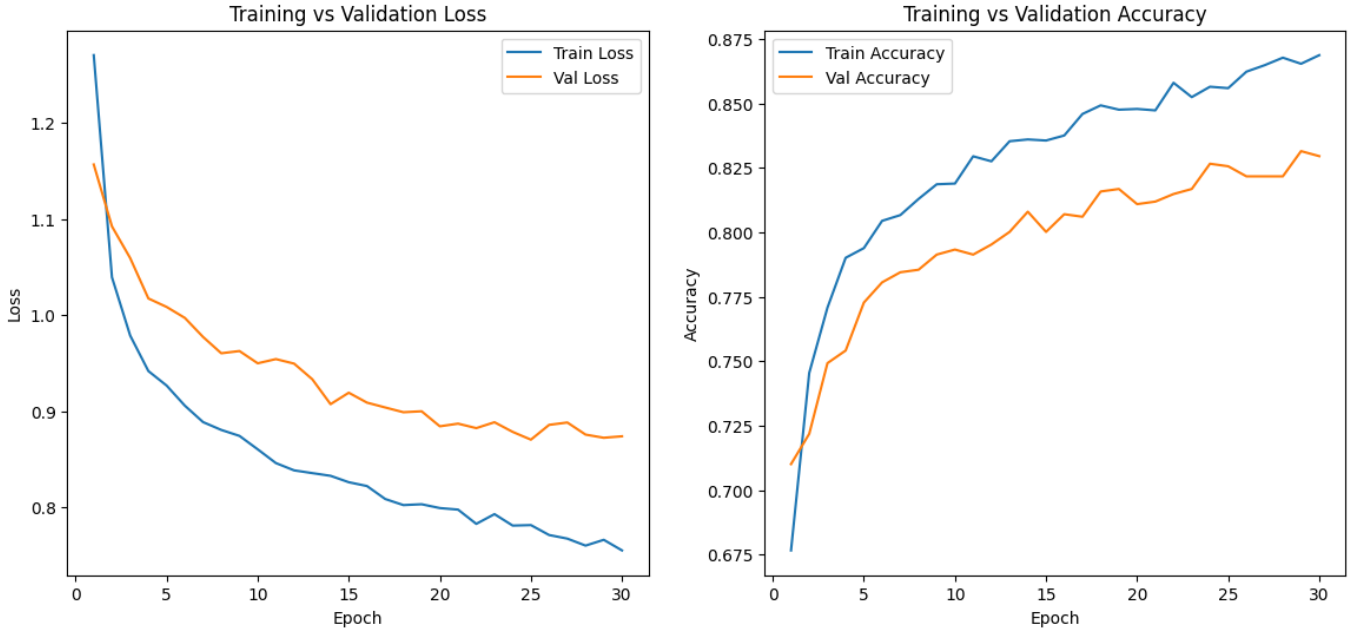


Fig 1: Plot of training and validation loss and accuracy, respectively

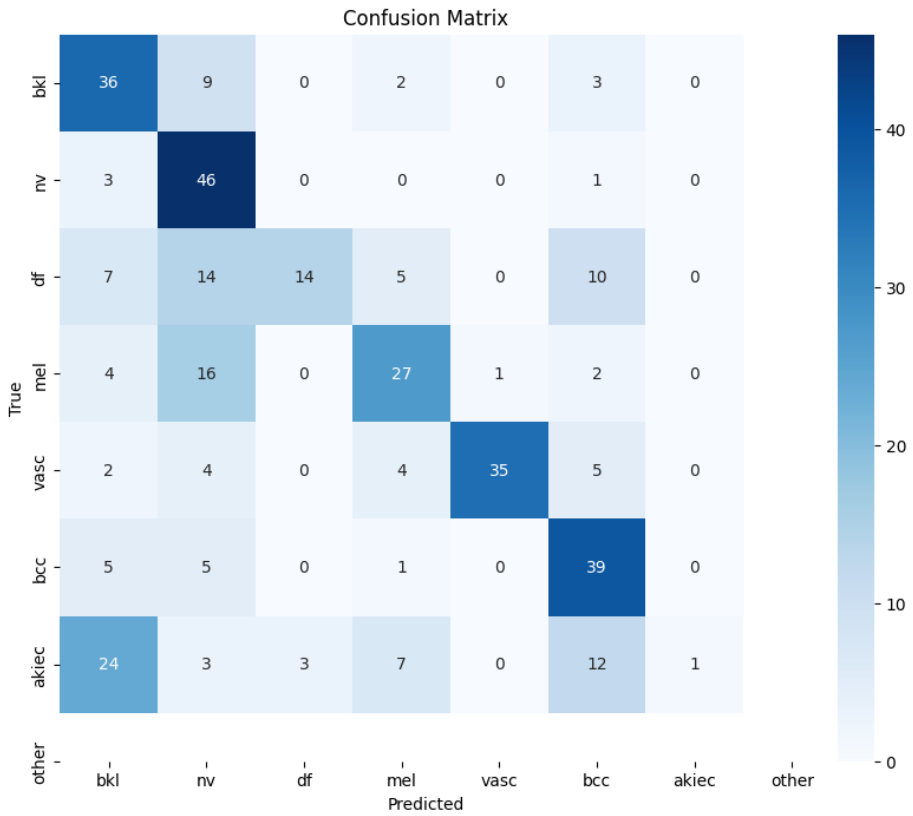


Fig 2: Plot of confusion matrix after testing on 350 test samples

## 5. **Learnings**

The project has given us a whole host of learnings which a text book course would not have. We have tried to summarise them as below:

a. Applying data augmentation on existing data in each training cycle to generalise better
b. Exploring better NN architectures to classify images. Actually the choice of architecture depends on data set size and purpose
c. Learnt about weighted classes for ensuring distribution of data uniformly. This is a very true phenomenon in real life.
d. Learnt visualisation of checking if overfitting is happening or not. This was only learnt in classrooms as part of theory but got to do it here
e. Used seaborn library on top of matplotlib to plot graphs easily without much code